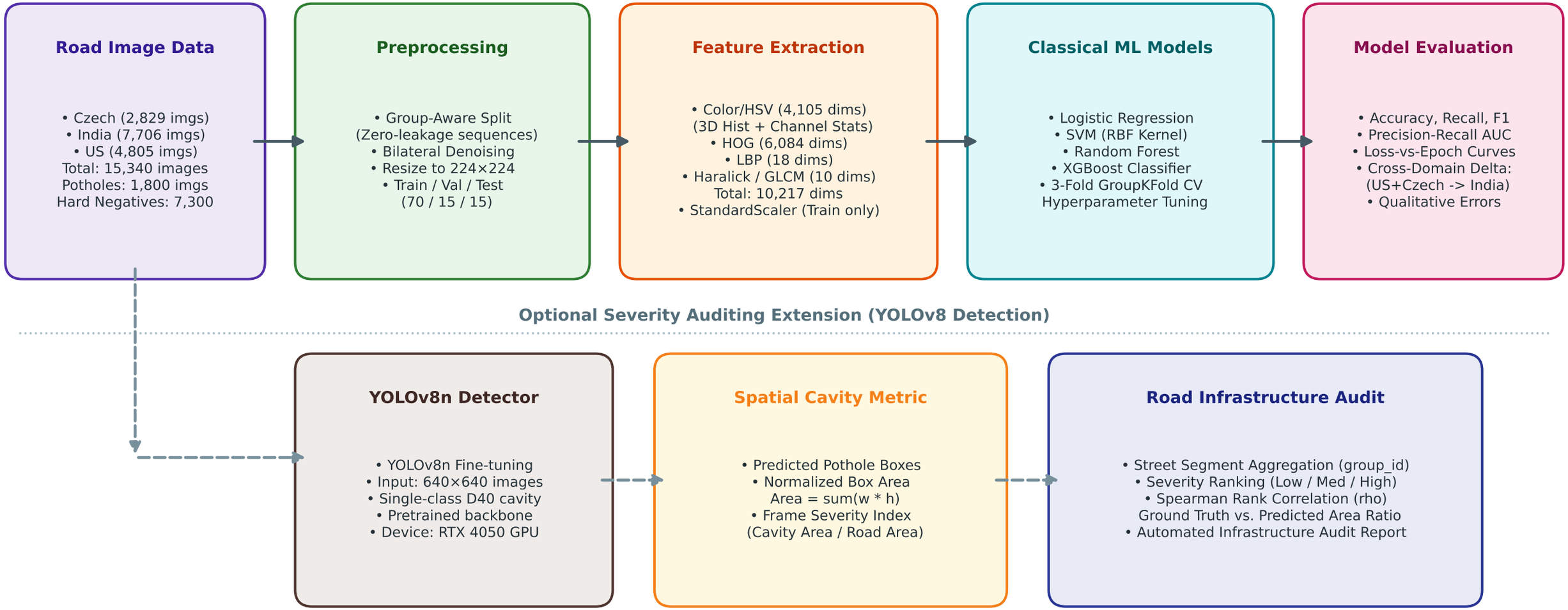
Road Image Data
• Czech (2,829 imgs)
• India (7,706 imgs)
• US (4,805 imgs)
Total: 15,340 images
Potholes: 1,800 imgs
Hard Negatives: 7,300
Preprocessing
• Group-Aware Split
(Zero-leakage sequences)
• Bilateral Denoising
• Resize to 224×224
• Train / Val / Test
(70 / 15 / 15)
Feature Extraction
• Color/HSV (4,105 dims)
(3D Hist + Channel Stats)
• HOG (6,084 dims)
• LBP (18 dims)
• Haralick / GLCM (10 dims)
Total: 10,217 dims
• StandardScaler (Train only)
Classical ML Models
• Logistic Regression
• SVM (RBF Kernel)
• Random Forest
• XGBoost Classifier
• 3-Fold GroupKFold CV
Hyperparameter Tuning
Model Evaluation
• Accuracy, Recall, F1
• Precision-Recall AUC
• Loss-vs-Epoch Curves
• Cross-Domain Delta:
(US+Czech -> India)
• Qualitative Errors
Optional Severity Auditing Extension (YOLOv8 Detection)
YOLOv8n Detector
• YOLOv8n Fine-tuning
• Input: 640×640 images
• Single-class D40 cavity
• Pretrained backbone
• Device: RTX 4050 GPU
Spatial Cavity Metric
• Predicted Pothole Boxes
• Normalized Box Area
Area = sum(w * h)
• Frame Severity Index
(Cavity Area / Road Area)
Road Infrastructure Audit
• Street Segment Aggregation (group_id)
• Severity Ranking (Low / Med / High)
• Spearman Rank Correlation (rho)
Ground Truth vs. Predicted Area Ratio
• Automated Infrastructure Audit Report